LESSON PLAN

ENGINEERING METROLOGY (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Power requirements, induced power, rotor, profile and parasite power and total power requirements.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Power requirements, induced power, rotor, profile and parasite power and total power requirements.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the helicopter control, main rotor movements, translation flight, power, auto rotation, vibration checks and various adjustments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of power requirement charts and identification of induced, profile, and parasite power components from IB diagrams.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three main components of total power required for helicopter flight?
(b) How does induced power change as airspeed increases?
(c) Define profile power in the context of rotor aerodynamics.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Total Power Requirements • Induced Power and Airspeed Relationship • Rotor Profile Power Characteristics • Parasite Power Components • Total Power Curve Analysis	L & Ds	PPT, IB	15 min
(b)	• Interaction of Power Components • Minimum Power Speed Analysis • Effect of Weight on Power • Effect of Density Altitude • Power Margins for Safety	L & Ds	PPT, PDF	20 min
(c)	• Power Requirements in Translation • Comparison with Fixed Wing • Impact on Flight Envelope • Operational Implications • Review of Power Definitions	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how induced power varies during the transition from hover to forward flight.
(b) Describe the relationship between parasite power and airspeed.
(c) Why is understanding total power requirements critical for flight safety?

12. Home Assignment and Follow Up	Prepare a brief note comparing the power requirements of a helicopter in hover versus translation flight, referencing the concepts covered in LP 02 regarding basic flight mechanics.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Auto rotation, ground resonance, vertex ring, blade sailing.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Subedar Major, P No. 12345	Course Coordinator, Major, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min